Shixin Zheng

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Appointments

Postdoctoral Associate August 2025 – Present
Department of Mathematics, University of Maryland, College Park, MD
Mentor: Haizhao Yang

Education

Purdue University, West Lafayette, IN
Ph.D. in Mathematics, September 2018 – August 2025 (degree conferred August 8, 2025)
Dissertation: "Riemannian Optimization Over PSD Fixed-Rank Matrix Constraints and Applications"
Advisor: Xiangxiong Zhang

Shandong University, Jinan, Shandong, China
B.Sc. in Mathematics and Applied Mathematics, September 2014 – May 2018

Research Interests

Computational optimization for machine learning and deep neural networks; Riemannian optimization and sampling on matrix manifolds; large-scale eigenvalue computation; convergence theory for neural-network training; scientific computing for the Gross–Pitaevskii eigenvalue problem.

Publications

Peer-reviewed journal articles

[1] S. Zheng, W. Huang, B. Vandereycken, and X. Zhang. "Riemannian optimization using three different metrics for Hermitian PSD fixed-rank constraints." *Computational Optimization and Applications*, 91(3):1135–1184, 2025. DOI: 10.1007/s10589-025-00687-8. arXiv:2204.07830.

[2] S. Zheng, H. Yang, and X. Zhang. "On the convergence of orthogonalization-free conjugate gradient method for extreme eigenvalues of Hermitian matrices: a Riemannian optimization interpretation." *Journal of Computational and Applied Mathematics*, 451:116053, 2024. DOI: 10.1016/j.cam.2024.116053. arXiv:2302.04974.

Preprints

[3] S. Zheng and X. Zhang. "Riemannian Hessian analysis of Sobolev gradient flows for the Gross–Pitaevskii ground state problem." arXiv:2610.12833, 2026.

[4] J. Li, S. Zheng, and X. Zhang. "Global convergence of an efficient splitting method for the defocusing Gross–Pitaevskii ground state problem." arXiv:2609.13509, 2026.

[5] S. Zheng, Y. Wang, and H. Yang. "Global Convergence and Error Propagation in Neural Gradient Flows: A Riemannian Optimization Framework." arXiv:2605.27779, 2026. Under review.

[6] T. Yu, S. Zheng, J. Lu, G. Menon, and X. Zhang. "Riemannian Langevin Monte Carlo schemes for sampling PSD matrices with fixed rank." arXiv:2309.04072, 2023.

Teaching

Instructor of Record, University of Maryland
STAT 400 / DATA 400 (Applied Probability and Statistics I), Fall 2025 and Fall 2026

Teaching Assistant, Purdue University 2018 – 2025
Recitation instructor, Calculus II; grader for Optimization, Numerical PDEs, Real Analysis, Linear Algebra, ODEs

Service and Presentations

- Reviewer: *SIAM Journal on Optimization*; *Journal of Computational Physics*; *Applied Mathematical Modelling*; *Journal of Fixed Point Theory and Applications*
- Co-organizer, Numerical Analysis Seminar, University of Maryland, 2025–26
- Invited speaker, *Mathematics of Dynamical Systems in Modern Machine Learning*, Brin Mathematics Research Center, University of Maryland, September 9–11, 2026
- Poster presentation, *New Trends in Scientific Computing*, IPAM, UCLA, April 2022

Honors and Awards

- T.T. Moh Scholarship, Department of Mathematics, Purdue University, 2018–19 and 2023–24

Last updated: October 2026.